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Title: Hyper-spectral & NVDI imaging of pathogen infection in Arabidopsis

Abstract:
Goal 1: To demonstrate that hyperspectral imaging can be used as a fast way to monitor plant pathogen infection.
Goal 2: to create a high-throughput hyperspectral imaging device

Introduction:
Purpose: To test if a bacterial titre in the shoot tissue of Arabidopsis seedlings causes a observable hyperspectral signature. This will test whether resistant varieties of plants have different hyperspectral responses to pathogens or FLG22.

We will be using an adapted protocol from this paper and add a hyperspectral imaging time series...
Ishiga Y, Ishiga T, Uppalapati SR, Mysore KS *(2011) Arabidopsis seedling flood-inoculation technique: a rapid and reliable assay for studying plant-bacterial interactions. Plant Methods* ***7****: 32*

"A flood-inoculation method that we have previously developed to infect the cotyledonary leaves of tomato [15] was modified to develop an Arabidopsis seedling flood-inoculation technique with reproducible disease symptoms. To perform uniform inoculation, 40 ml of bacterial suspension (OD_{600} = 0.4 in sterile water suspension) made in sterile distilled H2Ocontaining 0.025% Silwet L-77 (OSI Specialties Inc., Danbury, CT, U.S.A.) was dispensed into the plate containing 2-week-old Arabidopsis seedlings, and the plates were incubated for 2-3 min at room temperature. After the bacterial suspension was removed by decanta tion, plates containing inoculated plants were sealed with 3 M Micropore 2.5 cm surgical tape (3 M, St. Paul, MN, U.S.A.) and incubated at 24°C with a light intensity of 150-200 µEm-2 sec-1 and a 12 h light/12 h dark photoperiod. Symptom development was observed at 1 and 3 dpi. In each experiment, 16 plants were evaluated, and each experiment was repeated at least three times.
To determine the bacterial growth in Arabidopsis leaves, we measured internal bacterial population at several time points (0, 1, 2, 3 and 4 dpi). Internal bacterial populations were evaluated from four biological replicates and each replicate represented a pooled sample of four independent seedlings from a single experiment grown in a single Petri-dish. Inoculated seedlings were collected by cutting the hypocotyls to separate the above agar parts (whole rosette) from the Phytagel plate, and the total weight of inoculated seedlings was measured. After measurement of the seedlings' weight, the seedlings were surface-sterilized with 5% H2O2 for 3 min. After washing three times with sterile distilled water, a pooled sample of four seedling were homogenized in 10 mL sterile distilled water using a mortar and pestle, and diluted samples were plated onto MG or LB medium containing the appropriate antibiotics. Two days after plating of diluted samples, the bacterial colony forming units (CFU) were counted using proper diluted samples. The CFU was normalized as CFU/mg using total weight of inoculated seedlings. Bacterial populations were evaluated in three independent experiments."

Methods:

Genotypes: We will use 2 alleles of GLR3.3 (known to be susceptible to pathogen infection) that are susceptible to pathogen infection and comparing them to control Col-0.

Treatment: control vs FLG22 vs pathogenic bacterial infection

Experimental design:
4 petri dishes for each treatment (12 plates in total) ½ LS, 0.3%P
Each plate contains 3 rows (1 genotype per row) of plants grown on square horizontal plates.
The plates will be flooded with either a solution of FLG22, *Pseudomonas syringae* or Silwet for the control.
Plants will be imaged with a hyperspectral device before treatment and every 3 hours after treatment for 24 hours.

Results:
Assuming we can prove that there is a measurable difference in the hyperspectral response of resistant vs susceptible varieties then we will construct a robotic device to automate the movement of the hyperspectral imaging device.

Discussion and conclusions: